D. K.T. E. Society's

Textile and Engineering Institute, Ichalkaranji

(An Autonomous Institute)

AN AUTONOMOUS INSTITUTE

NAAC ACCREDITED WITH 'A +' GRADE

Syllabus

For

Final Year B. Tech.

Of

Computer Science and Engineering

(With effect from 2023-24)

DKTES Textile and Engineering Institute, Ichalkaranji

(An Autonomous Institute)

Teaching and evaluation Scheme for year 2023-24

Final Year B. Tech (Semester – VII) In Computer Science and Engineering

Sr. No.	Course Code	Course Title	Course Category	Teaching scheme				Course Credits	Evaluation scheme					
									Theory			Practical		TOTAL
									CIE		SEE	CIE	SEE	
				L	T	P	Contact Hrs/wk		MSE	ISE				
1	CSL441	Parallel Computing Architecture	PCC	3	-	-	3	3	30	20	50	-	-	100
2	CSL442	Data warehouse & Business Intelligence	PCC	3	-	-	3	3	30	20	50	-	-	100
3	CSLEL1	Elective-III	PEC	3	-	-	3	3	30	20	50	-	-	100
4	CSLEL2	Elective-IV	PEC	3	-	-	3	3	30	20	50	-	-	100
5	CSP449	Deep Learning Lab	PCC	2	-	2	4	3	-	-	-	50	50	100
6	CSP450	Data Warehouse and Business Intelligence Lab	PCC	-	-	2	2	1	-	-	-	50	50	100
8	CSPELII	Elective-IV Lab	PEC	-	-	2	2	1	-	-	-	50	-	50
9	CSD454	Project-I	PST	-	-	4	4	4	-	-	-	50	50	100
		Total		**14**	**-**	**10**	**24**	**21**	**120**	**80**	**200**	**200**	**150**	**750**

L- Lecture
T-Tutorial
P-Practical

SE-I : Semester Examination-I
SE-II : Semester Examination-II

CIE – Continuous In Semester Evaluation
SEE- Semester End Examination

CSLEL1 Elective-III	CSLEL2 Elective-IV	CSPEL2 Elective-IV Lab
CSL443 Advanced Machine Learning	CSL446 Internet of Things	CSP451 Internet of Things Lab
CSL444 Cyber Security	CSL447 Blockchain Technology	CSP452 Blockchain Technology Lab
CSL445 Big Data	CSL448 Computer Vision	CSP453 Computer Vision Lab

Course Category	HSMC (Hum. & Social Sc., Mgt)	BSC (Basic Sc.)	ESC Engg. Sc.)	PCC (Prof. Core Courses)	PEC (Prof. Elect. Courses)	OEC (Open Elct. Courses)	MC (Mandatory Courses)	PST (Project / Seminar / Ind. Training)
Credits	--	--	--	10	7	--	--	4
Cumulative Sum	3	19	22	73	7	3	--	5

Progressive Total Credits: 132+21 =153

DKTES Textile and Engineering Institute, Ichalkaranji

(An Autonomous Institute)

Teaching and evaluation Scheme for year 2023-24

Final Year B. Tech (Semester – VIII) In Computer Science and Engineering

Sr. No.	Course Code	Course Title	Course Category	Teaching scheme				Course Credits	Evaluation scheme					
									Theory			Practical		TOTAL
				L	T	P	Contact Hrs/wk		CIE		SEE	CIE	SEE	
									MSE	ISE				
1	CSL455	Feature Engineering	PCC	3	-	-	3	3	30	20	50	-	-	100
2	CSL456	Software Project Management	PCC	3	-	-	3	3	30	20	50	-	-	100
3	CSLEL1	Elective-III	PEC	3	-	-	3	3	30	20	50	-	-	100
4	CSLEL2	Elective-IV	PEC	3	-	-	3	3	30	20	50	-	-	100
5	CSP463	Full Stack Web Development Lab	PCC	2	-	2	4	3	-	-	-	50	50	100
6	CSP464	Mobile Application Development Lab	PCC	-	-	2	2	1	-	-	-	50	50	100
8	CSPELII	Elective-IV Lab	PEC	-	-	2	2	1	-	-	-	50	-	50
9	CSD468	Project-II	PST	-	-	8	8	8	-	-	-	50	50	100
		Total		**14**	**-**	**14**	**29**	**25**	**120**	**80**	**200**	**200**	**150**	**750**

L- Lecture
T-Tutorial
P-Practical

SE-I : Semester Examination-I
SE-II : Semester Examination-II

CIE – Continuous In Semester Evaluation
SEE- Semester End Examination

CSLEL 1 Elective-III	CSLEL 2 Elective-IV	CSPEL 2 Elective-IVLab
CSL457 Software Testing and Quality Assurance	CSL460 Software Development Framework	CSP465 Software Development Framework Lab
CSL458 Agile Methodology	CSL461 DevOps	CSP466 DevOps Lab
CSL459 Human Computer Interaction	CSL462 Quantum Computing	CSP467 Quantum Computing Lab

Course Category	HSMC (Hum. & Social Sc., Mgt)	BSC (Basic Sc.)	ESC Engg. Sc.)	PCC (Prof. Core Courses)	PEC (Prof. Elect. Courses)	OEC (Open Elct. Courses)	MC (Mandatory Courses)	PST (Project / Seminar / Ind. Training)
Credits	--	--	--	10	7	--	--	4
Cumulative Sum	3	19	22	73	7	3	--	5

Progressive Total Credits: 153+25 =178

DKTES Textile and Engineering Institute, Ichalkaranji

(An Autonomous Institute)

Teaching and evaluation Scheme for year 2023-24

Final Year B. Tech (Semester – VIII) In Computer Science and Engineering

Sr. No.	Course Code	Course Title	Course Category	Teaching scheme				Course Credits	Evaluation scheme					
									Theory			Practical		TOTAL
									CIE		SEE	CIE	SEE	
				L	T	P	Contact Hrs/wk		MSE	ISE				
1	CSL455	Feature Engineering	PCC	3	-	-	3	3	30	20	50	-	-	100
2	CSP463	Full Stack Web Development Lab	PCC	2	-	2	4	3	-	-	-	50	50	100
3	CSP464	Mobile Application Development Lab	PCC	-	-	2	2	1	-	-	-	50	50	100
4	CSD468	Project-II	PST	-	-	8	8	8	-	-	-	50	50	100
5	CST469	Internship	PST	-	-	10	10	10	-	-	-	50	50	100
		Total		**5**	**-**	**22**	**27**	**25**	**30**	**20**	**50**	**200**	**200**	**500**

L- Lecture
T-Tutorial
P-Practical

SE-I : Semester Examination-I
SE-II : Semester Examination-II

CIE – Continuous In Semester Evaluation
SEE- Semester End Examination

Course Category	HSMC (Hum. & Social Sc., Mgt)	BSC (Basic Sc.)	ESC Engg. Sc.)	PCC (Prof. Core Courses)	PEC (Prof. Elect. Courses)	OEC (Open Elct. Courses)	MC (Mandatory Courses)	PST (Project / Seminar / Ind. Training)
Credits	--	--	--	6	7	3	--	4
Cumulative Sum	3	23	22	68	14	17	--	9

Progressive Total Credits: 153+25 =178

Note: Feature Engineering, Full stack Web Development Lab, Mobile Application Development Lab and Internship will be conducted in online mode. Evaluation of these courses will be done by faculty allocated by the department in off-line mode.

DKTES Textile and Engineering Institute, Ichalkaranji
Third Year B. Tech. (Semester – VII)
CSL441: Parallel Computing Architecture

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs. /Week	03	MSE: 30 Marks
Tutorials: 00 Hrs./Week		ISE: 20 Marks
Practical: 00 Hrs./Week		SEE: 50 Marks

Pre-requisite: Computer Organization, Microprocessors, Microcontrollers

Course Outcomes:
On completion of the course, student will be able to–

1. Design parallel algorithms for different problems
2. Analyze performance of parallel programs
3. Describe shared memory architecture and cache coherence issues
4. Explain parallel algorithms for matrix multiplication, reduction and linear equations
5. Explain parallel algorithms for Sorting and searching.

Course Contents

Unit I	Introduction to Parallel Architecture	06 Hours

Introduction, Need of Parallel Architectures, classification of Architectures, Technology trends Multiprogramming, Time sharing, Multithreading, Multiprocessing concepts. Technology Tracks in parallel architectures.

Unit II	Pipeline Architectures	06 Hours

Principles of pipelining and array processing. Types of pipeline architectures, performance measures, Different hazards in pipeline architectures. Vector processing, Instruction sets, Cray type vector processor architecture.

Unit III	Shared Memory and Distributed Memory architectures	07 Hours

Shared memory multiprocessor models: UMA Model, NUMA model, COMA model, Distributed Memory Models: message passing multicomputer, shared memory programming with Open MP, Distributed Memory Programming with MPI.

Unit IV	Multithreaded and Dataflow Architectures	06 Hours

Latency-Hiding Techniques: Shared Virtual Memory, Prefetching Techniques, Distributed Coherent Caches. Principles of Multithreading, Scalable and Multithreaded Architectures. Dataflow Architectures: Concepts of data flow computing.

Unit V	Multiprocessor Architectures	06 Hours

Loosely coupled and tightly coupled architectures. Cluster computing as an application of loosely coupled architecture. Example –CM*, Inter cluster and Intra-cluster communication.

Unit VI	Parallel Programming	06 Hours

Parallel programming models, parallel languages and compilers, code optimization and scheduling, loop parallelization and pipelining.

Text Books:
1. Advanced computer architecture – Kai Hwang (MGH) 2. Computer Architecture and Parallel Processing – Kai Hwang and Briggs (MGH).
References Books:
1. Advanced Computer Architectures ,Parallelism, Scalability, Programmability, Kai Hwang & Naresh Jotwani, 2nd Edition McGrawHill Education.
Useful Links: --
1. https://www.slideshare.net/MahediMahfujAnik/parallel-searching **2. https://computing.llnl.gov/tutorials/parallel_comp/**

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSL442: Data warehouse and Business Intelligence

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs./Week	03	MSE: 30 Marks
Tutorials: 00 Hrs./Week		ISE: 20 Marks
Practicals: 00 Hrs./Week		SEE: 50 Marks

Pre-requisite: Database, Advanced Database

Course Outcomes:
On completion of the course, student should be able to–

1. Describe components and working of Data warehouse and Business Intelligence systems
2. Explain dimensional modelling using fact table and dimension table
3. Explain various components and working of ETL Systems
4. Explain analytical tools and reporting tools for Business Data Analysis

Course Contents

Unit I	Introducing the Technical Architecture:	08 Hours

Introduction: - The value of architecture, Technical Architecture overview, Back room Architecture, Presentation Server Architecture, Front room Architecture, Infrastructure, Metadata, Security.

Unit II	Introducing Dimensional Modelling:	07 Hours

Making the Case for Dimensional Modelling, Dimensional Modelling primer, Enterprise Data Warehouse Bus Architecture, More on Dimensions & Facts.

Unit III	Designing the Dimensional Modelling:	06 Hours

Modelling Process overview, Getting Organized, Four Step Modelling Process, Design the Dimensional Model.

Unit IV	Introducing Extract, Transformation & Load	06 Hours

Round up the requirements, the 34subsystems of ETL, Extracting Data, Cleaning & Conforming data, Delivering Data for Presentation

Unit V	Introducing Business Intelligence Applications	07 Hours

Importance of B.I. Applications, Analytical cycle for B.I, Types of B.I. Applications, Navigating Applications via the B.I portal

Unit VI	Designing & Developing B.I Applications	05 Hours

B.I. Application resource planning, B.I. Application Specification, B.I. Application Development, B.I. Application maintenance

Text Books:

1. Ralph Kimball, "The Data Warehouse Lifecycle toolkit', 2nd edition, Wiley India
References Books:
1. Data Warehousing: Fundamentals for IT Professionals by Paulraj Ponniah; 2nd Edn. Publisher: Wiley, John & Sons, Incorporated
2. Star Schema: The Complete Reference by Christopher Adamson, Mc-Graw Hill Osborne Media
3. The Data Warehouse Toolkit: The Complete Guide to Dimensional Modeling by Ralph Kimball
4. Data Warehousing in the Real World – Anahory & Murray, Pearson
5. https://nptel.ac.in/courses/106105174

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSL443: Advanced Machine Learning

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week	03	MSE: 30 Marks
Tutorials: 00 Hrs/Week		ISE: 20 Marks
Practicals: 00 Hrs/Week		SEE: 50 Marks

Course Outcomes:
On completion of the course, student should be able to–

1. Describe different classifier algorithm
2. Describe modern optimization techniques
3. Explain basics of association rule

Course Contents

Unit I	Introduction to SVM	07 Hours

Introduction to Support Vector Machine(SVM), Polynomial Support Vector Machines, Types of SVM Hyperplanes and Support Vectors, Working of SVM, Advantages and Disadvantages of SVM, Applications of SVM

Unit II	SVM Kernel	07 Hours

Dot product, Use of Dot product in SVM, Margin, Hard Margin SVM, Soft margin SVM, SVM Kernels, Types of Kernels, Parameters in Kernelized SVC, Computing the SVM classifier

Unit III	Ensemble Learning	06 Hours

Introduction, Bagging, Bagging ensemble family, Random Forest, Boosting, Boosting ensemble family, AdaBoost, Gradient Boosting Machines, XGBoost, Stacking Ensemble Learning, Stacked Generalization.

Unit IV	Introduction to Reinforcement Learning	08 Hours

Introduction, Terminology used in Reinforcement Learning, Evaluation aspect of reinforcement learning, Markov model for reinforcement learning, Q learning algorithm, Monte-carol algorithm.

Unit V	Optimization Techniques	07 Hours

Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Stochastic Gradient Descent (MB — SGD), SGD with Momentum, Nesterov Accelerated Gradient (NAG), Adaptive Gradient (AdaGrad), AdaDelta, Adam, Nadam

Unit VI	AutoML	06 Hours

Introduction, ML pipeline, Hyper-parameter tuning, AutoML, Challenges and limitation, Automated generation of end-to-end ML solutions, Automated image classification, End-to-end AutoML solutions Supervised ML Problems

Text Books:	
1	Kevin Murphy, Machine Learning: a Probabilistic Approach, MIT Press, 1st Edition, 2012, ISBN No.: 978-0262-30616-4
2	C.M. Bishop, Pattern Recognition and Machine learning, Springer, 1st Edition, 2013, ISBN No.: 978- 81-322-0906-5
3	Han, Jiawei Kamber, Micheline Pei and Jian, "Data Mining: Concepts and Techniques", Elsevier Publishers, ISBN:9780123814791, 9780123814807.
References Books:	
1	Peter Flach, Machine Learning: The Art and Science of Algorithms that make sense of data, Cambridge University Press, 1st Edition, 2012, ISBN No.: 978-1-316-50611-0 2.
2	Ethem Alpaydin, Introduction to Machine Learning, PHI, 2nd edition, 2013, 978-0-262-01243-0

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSL444: Cyber Security

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures:03 Hrs./Week Tutorials: 00 Hrs./Week Practicals: 00 Hrs./Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Computer Networking, Information Security

Course Outcomes:
On completion of the course, student should be able to–

1. Explain the cyber security concepts.
2. Describe the cyber security vulnerabilities and prevention techniques.
3. Explain the different rules and regulations under I.T. ACT.
4. Explain the concepts of digital forensics & incident management.

Course Contents

Unit I	Computer and Network Security	07 Hours

Introduction to Computer Security - Introduction, How Seriously Should You Take Threats to Network Security?, Identifying Types of Threats, Basic Security Terminology, Concepts and Approaches, Online Security Resources

Networks and the Internet : Introduction, Network Basics, How the Internet Works, Basic Network Utilities , Advanced Network Communications Topics,

Unit II	Cyber Frauds, DoS, Viruses	07 Hours

Cyber Stalking, Fraud, and Abuse: Introduction, How Internet Fraud Works, Identity Theft, Cyber Stalking, Protecting Yourself Against Cyber Crime.

Denial of Service Attacks: Introduction, DoS, Illustrating an Attack,Malware: Introduction, Viruses, Trojan Horses, The Buffer-Overflow Attack.The Sasser Virus/Buffer Overflow,Spyware, Other Forms of Malware, Detecting and Eliminating Viruses and Spyware,

Unit III	Techniques Used by Hackers	06 Hours

Introduction, Basic Terminology, The Reconnaissance Phase, Actual Attacks, Malware Creation, Penetration Testing

Unit IV	Computer Security Technology	06 Hours

Introduction, Virus Scanners, Firewalls, Antispyware, IDS, Digital Certificates, SSL/TLS, Virtual Private Networks,Wi-Fi Security,

Unit V	I.T. Act	05 Hours

Introduction, Cyber Security Regulations, Roles of International Law, the state and Private Sector in Cyberspace, Cyber Security Standards. The INDIAN Cyberspace, I.T. Act

Unit VI	Introduction to Forensics	05Hours

Introduction, General Guidelines, Finding Evidence on the PC, Finding Evidence in System Logs , Getting

Back Deleted Files, Operating System Utilities, Operating System Utilities, Mobile Forensics: Cell Phone Concepts
Text Books:
1. Computer Security Fundamentals - Chuck Easttom , Pearson ,third edition.
References Books:
1. Jason Luttgens, Matthew Pepe, Kevin Mandia, Incident Response & Computer Forensics, McGraw-Hill Osborne Media, 3 rd edition , 2014.
2. Keith J. Jones, Richard Bejtlich, Curtis W. Rose, Real Digital Forensics: Computer Security and Incident Response, Paperback – Import, 2005.
3 John Sammons, The Basics of Digital Forensics: The Primer for Getting Started in Digital Forensics
4. Hacking Exposed: Network Security Secrets & Solutions, Stuart McClure, Joel Scambray and George Kurtz, McGraw-Hill, 2005.
5. Network intrusion alert: an ethical hacking guide to intrusion detection, Ankit Fadia, Manu Zacharia, Thomson Course Technology PTR, 2007.
6. Ethical Hacking, Thomas Mathew, OSB Publisher, 2003.
7. Hacking Exposed: Network Security Secrets & Solutions, Stuart McClure, Joel Scambray and George Kurtz, McGraw-Hill, 2005.
8. Dave Shackleford, Virtualization Security: Protecting Virtualized Environments, John Wiley & Sons, 2012.
9. BRAGG, Network Security: The Complete Reference, McGraw Hill Professional, 2012

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSL445: Big Data

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs./Week Tutorials: 00 Hrs./Week Practicals: 00 Hrs./Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Database Engineering

Course Outcomes:

On completion of the course, student will be able to–

- Explore the fundamental concepts of big data and its analytics
- Analyze the big data using Hadoop and intelligent techniques
- Apply NoSQL big data management
- Recognize the suitable secure models for building competitive business decisions

Course Contents

Unit I	Importance of Big Data	06 Hours

Classification of Digital Data, Characteristics of Data, Evolution of Big Data, Definition of Big Data, Challenges with Big Data, What is Big Data? Other Characteristics of Data Which are not Definitional Traits of Big Data, Why Big Data? Traditional Business Intelligence (BI) versus Big Data, A Typical Data Warehouse Environment, A Typical Hadoop Environment, What is New Today? What is Changing in the Realms of Big Data? What is Big Data Analytics?, What Big Data Analytics isn't, Classification of Analysis, Challenges that prevent business from capitalizing Big Data, Top challenges facing Big Data, Why is Big Dataanalytics imp ortant?

Unit II	Hadoop Architecture	06 Hours

Hadoop ecosystem, Design of Hadoop distributed file system (HDFS), Anatomy of MapReduce job run, Classic Map-reduce, YARN, Failures in classic Map reduce and YARN, Data format, Analyzing data withHadoop, Scaling out, Hadoop streaming, Unit tests with MapReduce

Unit III	Hadoop I/O	06 Hours

Data Integrity - Data Integrity in HDFS, LocalFileSystem, ChecksumFileSystem, Compression – Codecs, Compression and Input Splits, Using Compression in MapReduce, Serialization - The Writable Interface, Writable Classes, Implementing a Custom Writable, Serialization Frameworks, Avro File-Based Data Structures – SequenceFile, MapFile

Unit IV	NoSQL Management	06 Hours

Introduction to NoSQL, Impedance mismatch, Emergence of NoSQL, Aggregate data models, Key-value and document data models, Column-family stores, Graph databases, Schemaless databases, Distribution models- sharding, Master-slave replication, Peer-peer replication, Sharding and replication Relaxing consistency – CAPTheorem, Relaxing durability

Unit V	Analytics Framework	06 Hours

Applications on Big Data Using Pig and Hive, Data processing operators in Pig, Hive services – HiveQL, Querying Data in Hive, Fundamentals of HBase and ZooKeeper

Unit VI	Securing Ecosystem	06 Hours

Steps to secure big data, Classifying Data, Protecting–Big Data, Configuring Kerberos for Hadoop, Securingecosystem components – Pig, Hive, Oozie, Flume

Text Books:

1. Michael Mineli, Michele Chambers, Ambiga Dhiraj, “Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses” (Wiley Publication)
2. Tom White, “Hadoop: The Definitive Guide” (O’Reilly Media)
3. P. J. Sadalage, M. Flower, “NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence” (Addison-Wesley)
4. Sudeesh Narayanan, “Securing Hadoop” (O’Reilly Media)

References Books:

1. Seema Acharya, Subhasini Chellappan, “Big Data Analytics”, Wiley.
2. Chris Eaton,Dirk derooset al., “Understanding Big data”, McGraw Hill.
3. G James, D. Witten, T Hastie, R. Tibshirani, “An Introduction to Statistical Learning: with Applications in R”, Springer.
4. Douglas Eadline, “Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem”, Pearson Education.
5. E. Capriolo, D. Wampler, J. Rutherglen, “Programming Hive”, O’ Reilly.
6. Lars George, “HBase: The Definitive Guide”, O’ Reilly.
7. Alan Gates, "Programming Pig”, O’ Reilly

Useful Links:

1. Analytics Vidhya (http://www.analyticsvidhya.com/) ...
2. Dataversity (http://www.dataversity.net/) ...
3. R Bloggers (http://www.r-bloggers.com/) ...
4. SmartData Collective (http://www.smartdatacollective.com/) ...
5. Data Science Central (http://www.datasciencecentral.com/) ...
6. Planet Big Data (http://planetbigdata.com/)

DKTE Society's Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSL446: Internet of Things

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs. /Week Tutorials: 00 Hrs./Week Practicals: 00 Hrs./Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Networking

Course Outcomes:

On completion of the course, student should be able to–

1. Explain the fundamental concepts and principles of IoT.
2. Describe the architecture and components of IoT systems.
3. Describe basics of Embedded Devices.

Course Contents

Unit I	Internet of things Overview	07 Hours

What is the Internet of Things? About objects/things in the IoT, The identifier in the IoT , Examples of applications, Enabling technologies of IoT, Identification technology, Sensing and actuating technology, Other technologies, Connected objects' communication.

Unit II	RFID and IoT Architecture	07 Hours

Introduction, Principles of RFID, Components of an RFID system, Layered Architecture for IoT, Protocol Architecture of IoT, Three layers and five layers Architecture of IoT, Fog architecture of a smart IoT gateway

Unit III	Network Protocols	05 Hours

Introduction to ZigBee, BLE, WiFi, LTE, IEEE 802.11ah, Discuss data rate, range, power, computations/bandwidth, QoS.

Unit IV	Data Protocols	06 Hours

Message Queue Telemetry Transport, Constrained Application Protocol, Advanced Message Queuing Protocol Data Distributed Services, Representational State Transfer, Extensible Messaging and Presence Protocol.

Unit V	Application Development	06 Hours

Programming languages for IoT (e.g., Python, C), Development frameworks and tools (e.g., Arduino, Raspberry Pi, Node-RED), Web and mobile app development for IoT

Unit VI	Deployment and Case Studies	05 Hours

Challenges in deploying IoT systems, Industry and real-world IoT implementations, Smart cities, healthcare, agriculture, and industrial IoT

Text Books:
1. The Internet of Things: Connecting Objects to the Web, Hakima Chaouchi, Wiley 2. Interconnecting Smart Objects with IP: The Next Internet, Jean-Philippe Vasseur, Adam Dunkels, Morgan Kuffmann 3. Designing the Internet of Things , Adrian McEwen (Author), Hakim Cassimally
References Books:
1. Interconnecting Smart Objects with IP: The Next Internet, Jean-Philippe Vasseur, Adam Dunkels, Morgan Kauffmann, Published by Elsevier.
2. Internet of Things, by Srinivasa K.G.,Siddesh G.M.,HanumanthaRaju R.,1st Edition, CENGAGE Leaning India, 2017.

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSL447: Blockchain Technology

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures:03 Hrs./Week Tutorials: 00 Hrs./Week Practicals: 00 Hrs./Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Computer Network, Information Security

Course Outcomes:
On completion of the course, student should be able to–

1. Describe basics concepts of blockchain.
2. Apply blockchain concepts to secure data.
3. Analyze different cryptocurrency.

Course Contents

Unit I	Introduction to Blockchain	06 Hours

Basics of Blockchain:What is Blockchain?,Who uses Blockchain?,Need of Blockchain,-Distributed Vs Centrallized Vs Decentralized -Public Ledgers: Blockchain as Public Ledgers, Blockchain Distributed Ledger-Applications of Blockchain-Evolution of Blockchain (Blockchain Version)-History of Blockchain-Types of Blockchain,-Benifites and Challenges of Blockchain Usages–Limitations of Blockchain-Blockchain Vs Database

Unit II	Architecture Design	07 Hours

Blockchain Design Principles - Components of blockchain - Layered Architecture of Blockchain Ecosystem - Essentials of Blockchain : Blockchain Generation -Types of Networks – Distributed Network P2P Network -Blockchain Consensus Protocols: Proof of Work, Proof of Stake

Unit III	Working of Blockchain	06 Hours

Structure of Blockchain-Basics of Blockchain Development-prerequisite for the blockchain development - Blockchain Development Tools-Top blockchain platforms- How Blockchain Works? - Understanding SHA256 Hash- Immutable Ledger- Distributed P2P Network- How Mining Works? (The NONCE and Cryptographic Puzzle)- Byzantine Fault Tolerance- Consensus Protocols[Proof of Work, Proof of Stake, Defense Against Attackers, Competing Chains]- Consensus Mechanisms-Blockchain Demo

Unit IV	Introduction to web3.js	06 Hours

Importing web3.js,Connecting to nodes, The API structure, BigNumber.js, Unit conversion, Retrieving gas price, balance, and transaction details, Sending ether, Working with contracts, Retrieving and listening to contract events, Building a client for an ownership contract, The project structure, Building the backend, Building the frontend, Testing the client

Unit V	Blockchain with Cryptocurrency- Bitcoin & Ethereum	06 Hours

Introduction of Cryptocurrency- Basics of Cryptocurrency -Advantages as well as Disadvantages of the Cryptocurrencies -Bitcoin – Introduction- Bitcoin Working- Transactions -Buy Bitcoin -Bitcoin Mining-

Value of Bitcoin -Wallet - Blocks - Merkley Tree -Bitcoin with Blockchain- Ethereum- Introduction- Components of Ethereum- DApp in Ethereum- Digital Tokens		
Unit VI	**Smart Contracts**	**06 Hours**
Introduction-What is Smart Contract?-How do smart contract work?- Ethereum Virtual Machine, Ether,Gas - Ethereum and Smart Contracts-Wallets for Ethereum - Ethereum Network - Ethereum Smart Contracts -DApps -Decentralized Autonomous Organizations[DAO] -Hard and Soft Forks- Initial Coin Offerings- Demo of Smart Contracts		

Text Books:

1. Beginning Blockchain:A Beginner's Guide to Building Blockchain Solutions By Bikramaditya Singhal,
2. Gautam Dhameja, Priyansu Sekhar Panda, Apress Media
3. Melanie Swan, "BlockChain: Blueprint for a New Economy", O"Reilly, first edition – 2015.
4. Daniel Drescher, "BlockChain Basics", Apress; 1stedition, 2017
5. Anshul Kaushik, "BlockChain and Crypto Currencies", Khanna Publishing House, Delhi.
6. Imran Bashir, "Mastering BlockChain: Distributed Ledger Technology, Decentralization and Smart
7. Contracts Explained", Packt Publishing, first edition – 2012

References Books:

1. Ritesh Modi,"Solidity Programming Essentials: A Beginner's Guide to Build Smart Contracts for Ethereum and BlockChain", Packt Publishing.
2. Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guild to Blockchain Technology and
   Blockchain Programming', Create Space Independent Publishing Platform, 2017
3. Mastering Blockchain by Imran Bashir, Third Edition, Packt Publication 3. Waterhole, The Science of the Blockchain
4. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System
5. Mastering Ethereum: Building Smart Contracts and DAPPS, by Andreas Antonopoulos, Dr. Gavid Wood, Oreilly Publication

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. CSE (Semester – VII)
CSL448: Computer Vision

Teaching Scheme:	Credits 03	Evaluation Scheme:
Lectures: 03 Hrs. /Week		MSE-I: 30 Marks
Tutorials: 00 Hrs./Week		ISE-II: 20 Marks
Practical: 00 Hrs./Week		SEE: 50 Marks

Course Outcomes:

On completion of the course, Students will be able o –

1. Explain concepts applied in computer vision.
2. Apply spatial and geometric transformation on image.
3. Design solution for the given computer vision problem

Course Contents

Unit I	Introduction	07 Hours

Introduction to Computer Vision, Computer Vision Applications, Digital image Representation, , Image Model, Image sampling and Quantization, grayscale resolution, pixel relationship, Basic grey level transformations, image negation, log transformations, power law transformations, piece-wise linear transformations, histogram processing, Image enhancement using arithmetic and logical operations, neighborhood operations, Color Image Models.

Unit II	Feature Extraction	07 Hours

Spatial feature Extraction, transform feature, Edge detection, Boundary Representation, Region Representation and Moment Representation, Structure-Shape Features, Texture, Scene Matching and Detection-Image Segmentation

Unit III	Restoration and Compression	06 Hours

Image restoration, Degradation Model, Discrete Formulation, Restoration in the Presence of Noise Only - Spatial Filtering.
Compression Fundamentals – Compression Models – Error Free Compression – Lossy Compression

Unit IV	Transformation	07 Hours

Geometric transformations (rotation, scaling, etc.), Image warping and homography, Interest point detection (Harris corner detector, FAST, etc.), Feature descriptors

Unit V	Object Recognition	06 Hours

Patterns and Pattern Classes, Recognition Based on Decision-Theoretic Methods, Matching, Optimum Statistical Classifiers, Structural Methods, Matching Shape Numbers, String Matching. machine learning-based methods.

Unit VI	Applications of Computer Vision	06 Hours

Recognition of characters, Fingerprints and faces, image classification -Image databases, How the 3rd dimension changes the problem. Stereo 3D description, 3D model.

Text Books:

1. Fundamentals of Digital Image Processing- A.K.Jain (PHI)
2. Image Processing and Machine vision-Milan Sonka, Vaclav Hlavae, Roger Boyle Cengage Learning India Pvt Ltd (2008)
3. R. C. Gonzalez, R. E. Woods, Digital Image Processing, 4th Edition. 2018, PHI

References Books:

1. Boyle R & Thomas R, Computer Vision – A First Course, 2nd Edition, McGraw Hill, 1990.
2. Computer Vision: A Modern Approach" by David A. Forsyth and Jean Ponce
3. Computer Vision: Algorithms and Applications" by Richard Szeliski

Useful Links: --

1. **https://machinelearningmastery.com/what-is-computer-vision/**

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSP449: Deep Learning Lab

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 02 Hrs./Week Tutorials: 00Hrs./Week Practicals: 02 Hrs./Week	03	CIE: 50 Marks SEE: 50 Marks

Pre-requisite: Machine Learning, Advanced Machine Learning.

Course Outcomes:
On completion of the course, student should be able to–

1. Explain basic concepts of deep neural network.
2. Construct deep learning networks for given task.
3. Compare different deep learning networks on various evaluation parameters.

Course Contents

Unit I	Introduction to Deep Learning	07 Hours

Relation between artificial intelligence, machine learning, deep learning, Introduction to neural network, Data representations for neural networks, Tensor Operations, Gradient based optimization, Tensorflow, Keras, Brief history of Tensorflow and Keras, Setting up deep learning workspace, Working with Tensorflow, Understanding core Keras API, Examples: Classification of movie review, Classification of newswires, House price prediction systems.

Unit II	Developing Models	06 Hours

Different ways to develop Keras model: Sequential model, Functional API, Subclassing, Using built-in training and evaluation loops, developing own training and evaluation loop.

Unit III	Deep learning for Computer Vision	06 Hours

Introduction to ConvNet, Training convent from scratch on a small dataset, Leveraging pre-trained model, Three essential computer vision tasks, Image segmentation, Modern convent architecture patterns, Interpreting what convnets learn.

Unit IV	Deep learning for time-series data	06 Hours

Different kinds of time series data, A temperature forecasting example, Recurrent neural network, Advanced use of recurrent neural networks

Unit V	Deep learning for Text	07 Hours

Natural language processing, Pre-paring text data, Two approaches: sets and sequence, Transformer architecture, Sequence to sequence learning

Unit VI	Best practices for real word	07 Hours

Hyper-parameter optimization, model ensemble, Speed up training on GPU with mixed precision,

Multi-GPU training, TPU training.

Text Books:

1. Deep Learning with Python, 2nd Edition - François Chollet (Manning Publications).

References Books:

1. Deep learning - Ian Goodfellow and Yoshua Bengio and Aaron Courville (MIT Press)

Important Links

1. https://github.com/fchollet/deep-learning-with-python-notebooks

Experiment List

1. Installing and working with Tensorflow and Keras with Anaconda or Miniconda
2. Introduction and working with Google Colab for using GPUs and TPUs for large projects
3. Design and develop a model for predicting house prices.
4. Design and develop shallow neural network for classifying hand written digit recognition.
5. Design and develop deep neural network for classifying hand written digit recognition.
6. Design and develop CNN for classifying images.
7. Develop a CNN model with transfer learning for classifying images.
8. Develop a CNN model to perform image segmentation.
9. Develop a CNN model with residual block.
10. Design and develop Xception-like model.
11. Visualizing convnet filters.
12. Identify sentiment of given text using unigram / bigram with TF-IDF encoding
13. Design and develop RNN model for temperature forecasting
14. Design and develop GRU/LSTM model for sentiment analysis.
15. Hyper-parameters tuning for getting better result
16. Design and develop a deep learning model for handling multiple input and multiple output
17. Handling callbacks of deep neural network.
18. Monitoring and visualization of model with tensorboard
19. Design and develop a transformer based model for text classification.
20. Design and develop a model for transferring source language to destination language.

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSP450: Data warehouse and Business Intelligence Lab

Lab Scheme: Practical: 02 Hrs. /Week	Credits 01	Evaluation Scheme: CIE: 50 Marks SEE: 50 Marks

Course Outcomes:

On completion of the course, student will be able to–

1. Install and configure BI Systems.
2. Design dimensional modelling using fact table and dimension table.
3. Design ETL process
4. Apply analytical and reporting tools for various BI tasks.

List of Experiments
(It should consist of 10-12 experiments based on the following topics.)

1	Install and configure BI Systems
2	Design fact and dimension tables for specific applications
3	Dimensional Modeling and Schema Design for given business process
4	Data profiling and pre-processing of data before Extract Transfer Load
5	Data Cleansing and Deduplication using suitable tools
6	Demonstration of ETL using suitable ETL tools
7	Demonstration of slowly changing dimensions using suitable tools
8	Predictive Analytics and Data Mining
9	Data Visualization and Reporting
10	Data Governance and Metadata Management
11	Creating Dashboards using suitable tools
12	Creating Scorecards using suitable tools

DKTE Society's Textile and Engineering Institute, Ichalkaranji Final Year B. Tech. (Semester – VII) CSP451: Internet of Things Lab		
Teaching Scheme: Lectures: 00 Hrs /Week Tutorials: 00 Hrs/Week Practicals: 02 Hrs/Week	Credits 01	Evaluation Scheme: CIE: 50 Marks SEE: 00 Marks

Course Outcomes:

On completion of the course, student should be able to–

1. Understand the concept of Internet of Things.
2. Implement interfacing of various sensors with Arduino/Raspberry Pi.
3. Demonstrate Show an ability to upload/download sensor data on cloud and server.
4. Show the ability to transmit data wirelessly between different devices.

Course Contents

List of Experiments **(It should consist of 10-12 experiments based on the following topics.)**	
1	Introduction to IoT enabled Hardware platforms
2	LED blinking using Node MCU and cloud
3	LED blinking using Raspberry pi and Cloud
4	Sine wave generation on cloud using raspberry pi
5	Interfacing of HC05 Bluetooth sensor and data logging on Android smart phone
6	Ultrasonic sensor SR 04 interfacing with Node MCU and uploading data on cloud
7	Ultrasonic sensor SR 04 interfacing with Raspberry pi and uploading data on cloud
8	Arduino Uno and LED interfacing and controlling through Blynk cloud
9	XBee communication – Router, Co-Ordinator and end node.
10	Own cloud configuration on raspberry pi- create local server and control different led's
11	Smart home application using any hardware platform.
12	E-PUC application using any hardware platform.

DKTES Textile and Engineering Institute, Ichalkaranji Final Year B. Tech. (Semester – VII) CSP452: Blockchain Technology Lab		
Lab Scheme: Practical: 02 Hrs. /Week	Credits 01	Evaluation Scheme: CIE: 50 Marks SEE: - Marks

Course Outcomes:

On completion of the course, student will be able to–

1. Describe basic concepts of blockchain technology
2. Implement basic concepts of blockchain.

List of Experiments

1	Write a solidity program that takes positive integer n as input and returns the integer obtained by reversing the digits in n.
2	Write a solidity program to sum all prime numbers from 1 to 300.
3	Write a smart contract to Store and display Information of Student.
4	Design a contract to represent bank account. Include the following members: State variables: Name of the depositor, Account number, Type of account, Balance amount in the account Methods: • To deposit an amount • To withdraw an amount after checking the balance • To display the name and balance
5	Design a web3.js application for Information of Student.
6	Design a web.js application for bank account.
7	Design a web3.js application for the lottery application
8	Design a web3.js application for property transfer application

DKTES Textile and Engineering Institute, Ichalkaranji.
Final Year B. Tech. (Semester – VII)
CSP453: Computer Vision Lab

Lab Scheme:	Credits	Evaluation Scheme:
Practical: 02 Hrs. /Week	01	CIE: 50 Marks

Course Outcomes:

On completion of the course, students will be able to–

1. Illustrate the image fundamentals, and mathematical transforms necessary for image processing.
2. Apply the various techniques of image enhancement.
3. Develop a mini project by applying various methodologies of computer vision.

List of Experiments
(It should consist of 10-12 experiments based on the following topics.)

1	Elementary Image Handling and Processing Operations on the Image.
2	Image Arithmetic operations on the image.
3	Point operations on the image.
4	Neighborhood operations on the image.
5	Image Histogram.
6	Image segmentation.
7	Image Filtering.
8	Segmentation based on Discontinuities.
9	Image Edge detection.
10	Mini Project-based on Computer Vision Applications.
11	Study of image compression standards.
12	Study of computer vision libraries. .

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VII)
CSD454: Project-I

Lab Scheme:	Credits	Evaluation Scheme:
Practical: 04 Hrs. /Week	04	CIE: 50 Marks SEE: 50 Marks

Pre-requisite: Software Engineering

Course Outcomes:

On completion of the course, a team of student will be able to–

1. Define problem statement for selected project topic
2. Analyze the requirements of the selected project topic
3. Develop the design for providing solution to the selected problem statement.
4. Write a report covering SRS and design of the project
5. Give presentation on project.

Project Guidelines

In order to form a project group, students are required to assemble teams comprising four to five individuals. Each group must submit their area of interest, along with the relevant student information, using the provided form. Subsequently, based on the indicated area of interest and the expertise of the faculty, each group will be assigned a guide. It is necessary for the students to consult with their respective guides in order to select a suitable project topic. The finalization of the chosen topic will be conducted by the project evaluation team. Once the topic has been selected and finalized, the group must proceed with the preparation and presentation of the project's synopsis. This synopsis should encompass an introduction, related work, project topic selection including the problem statement, objectives, and methodology. This presentation will serve as the first presentation of the project-I. In the second presentation, the group is required to develop a software requirement specification (SRS) document. The evaluation of the SRS will be based on various criteria, including requirement gathering, analysis, teamwork, and presentation skills. Furthermore, during the internal exam, the evaluation will be focused on the design and report of the project. Evaluation of internal exam is based on various criteria like tools, environmental factors, ethics, cost, self-learning, presentation, and team work.

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VIII)
CSL455: Feature Engineering

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs./Week Tutorials: 00Hrs./Week Practicals: 00 Hrs./Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Data Science , Machine Learning

Course Outcomes:

On completion of the course, student will be able to–

1. Explain basics of feature engineering used for representing and generating process.
2. Describe features of different types of data with feature selection process.
3. Describe feature transformations process for converting high dimensional features to low dimensional features.
4. Explain feature learning process from the given input.

Course Contents

Unit I	Introduction to feature engineering	06 Hours

Motivating example – AI-powered communications, importance of feature engineering, introduction of feature engineering, Evaluation of machine learning algorithms and feature engineering procedures, Feature understanding, Feature improvement – cleaning datasets, Feature selection – removing bad attributes, Feature construction, Feature transformation, Feature learning

Unit II	Basics of Feature Representation	06 Hours

Scalars, Vectors, and Spaces, Dealing with Counts, Binarization, Quantization or Binning, Log Transformation, Feature Scaling or Normalization, Min-Max Scaling, Standardization (Variance Scaling), ℓ2 Normalization, Interaction Features, Feature Selection

Unit III	Features of Text and Categorical Data	08 Hours

Bag-of-X: Turning Natural Text into Flat Vectors, Filtering for Cleaner Features, Atoms of Meaning: From Words to n-Grams to Phrases, Tf-Idf : A Simple Twist on Bag-of-Words, Putting It to the Test, Deep Dive, Encoding Categorical Variables, Dealing with Large Categorical Variables

Unit IV	Feature Selection	06 Hours

Importance of Feature Selection in Machine Learning, Goals of Feature Selection, Classes of Feature Selection Methodologies, Effect of Irrelevant Feature, Overfitting to Predictors and External Validation, Greedy Search Methods- Simple Filters, Recursive Feature Elimination, Stepwise Selection

Unit V	Feature Transformations	08 Hours

Intuition, Derivation, Linear Projection, Variance and Empirical Variance, Principal Components: First Formulation, Principal Components: Matrix-Vector Formulation, General Solution of the Principal Components, Transforming Features, Implementing PCA, PCA in Action, Whitening and ZCA, Considerations and Limitations of PCA, Use Cases

Unit VI	Feature Learning	06 Hours
Parametric assumptions of data, Non-parametric fallacy, feature learning algorithms, Reconstructing the data, The Bernoulli RBM, Extracting PCA components from MNIST, Extracting RBM components from MNIST, Using RBMs in a machine learning pipeline, Learning text features – word vectorizations, Word embeddings, Application of word embeddings – information Retrieval		
Text Books:		
1. Sinan Ozdemir, Divya Susarla, "Feature Engineering Made Easy", Packt Publishing, ISBN 978-1-78728-760-0 2. Alice Zheng & Amanda Casari, "Feature Engineering for Machine Learning: Principles and Techniques for data scientist", Oreilly		
References Books:		
1. Max Kuhn , Kjell Johnson, "Feature Engineering and Selection: A Practical Approach for Predictive Models" 1st Edition, Chapman & Hall/CRC Data Science Series, ISBN 13-978-1-138-07922-9		
Useful Links:		
1. https://machinelearningmastery.com/discover-feature-engineering-how-to-engineer-features-and-how-to-get-good-at-it/		

DKTES Textile and Engineering Institute, Ichalkaranji.
Final Year B. Tech. (Semester – VIII)
CSL456: Software Project Management

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week	03	MSE: 30 Marks
Tutorials: 00Hrs/Week		ISE: 20 Marks
Practicals: 00 Hrs/Week		SEE: 50 Marks

Pre-requisite: Software Engineering

Course Outcomes :
On completion of the course, students will be able to

1. Describe project management methodology and documents used in project management at each stage of the project life cycle.
2. Develop the scope of work, Work Breakdown structure, a Gantt chart, and milestones of the project.
3. Appreciate the elements of quality, human resources, and risk in projects.
4. Acquaint themselves with project cost and cost management concepts.

Course Contents

Unit I	Introduction to S/W Project Management	07 Hours

Project and Project Management (PM), Role of Project Manager, System view of PM, Organization, Stakeholders, Project phases and lifecycle, Context of IT projects, process groups, mapping groups to Knowledge area.

Unit II	Project Integration Management	06 Hours

Strategic planning and project selection, Developing a Project Management Plan, Directing and Managing Project Work, Monitoring and Controlling Project Work, Performing Integrated Change Control, and Closing Projects or Phases.

Unit III	Project Scope, Time and Cost Management	06 Hours

Planning Scope Management, Collecting Requirements, Defining Scope, Creating the Work Breakdown Structure, Validating Scope, Controlling Scope, Planning Schedule Management, Defining Activities, Sequencing and Estimating Activity, Resources & Duration, Developing & Controlling Schedule Basic Principles of Cost Management, Planning Cost Management, Estimating Costs, Determining the Budget, Controlling Costs.

Unit IV	Quality Management	06 Hours

Importance, Planning Quality Management, Performing Quality Assurance, Controlling Quality, Tools and Techniques for Quality Control, Human Resource Management: Importance, keys to managing people, human resource planning, acquiring, developing, and managing the project team.

Unit V	Human Resource Management	06 Hours

Human Resource Management: Importance, keys to managing people, human resource planning, acquiring, developing, and managing the project team.

Unit VI	Risk management	08 Hours

Importance, risk management planning, sources of risk, risk identification, qualitative and quantitative risk analysis, risk response planning, risk monitoring and control.

Text Books:
1. Information Technology Project Management, Kathy Schwalbe, Cengage Learning 7E
References Books:
1. Project Management Core Textbook – Mantel Jr., Meredith, Shafer, Sutton with Gopalan (Wiley India Edition
2. Financial Management: Text and Problems by M Y Khan & P K Jain, Publisher: TMH, New Delhi

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VIII)
CSL457: Software Testing and Quality Assurance

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week Tutorials: 00Hrs/Week Practicals: 00 Hrs/Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Course Outcomes:

On completion of the course, student should be able to–

1. Explain the fundamentals of software testing, quality assurance and software testing methods.
2. Illustrate the software verification and validation process for different software engineering artifacts.
3. Design the different test cases for software testing

Course Contents

Unit I	Introduction to Software Testing	06 Hours

Introduction: Introduction, Testing Goals of Software Testing, Software Testing Definitions, Model for Software Testing, Effective Software Testing vs. Exhaustive Software Testing, Software Testing as a Process, Software Failure Case Studies.

Software Testing Terminology and Methodology: Software Testing Terminology, Software Testing Life Cycle (STLC), Software Testing Methodology

Unit II	Verification and Validation	08 Hours

Software Verification: Verification and Validation (V&V) Activities, Verification, Verification of Requirements, Verification of High-level Design, Verification of Low-level Design, Verification of source code

Static Testing: Inspections, Structured Walkthroughs, Technical Reviews

Validation Activities: Unit Validation Testing, Integration Testing, Function Testing, System Testing, Acceptance Testing.

Unit III	Regression Testing and Software Metrics	07 Hours

Regression Testing: Progressive vs. Regressive Testing, Regression Testing Produces Quality Software, Regression Testability, Objectives of Regression Testing, Regression Testing Types, Regression Testing Techniques

Software Metrics: Need of Software Measurement, Definition of Software Metrics, Classification of Software Metrics, Entities to be Measured, Size Metrics.

Unit IV	Dynamic Testing: Black-Box Testing Techniques	06 Hours

Boundary Value Analysis (BVA), Equivalence Class Testing, State Table-Based Testing, Decision Table-Based Testing, Cause-Effect Graphing Based Testing, Error Guessing

Unit V	Dynamic Testing: White-Box Testing Techniques	07 Hours

Need of White-Box Testing, Logic Coverage Criteria, Basis Path Testing, Graph Matrices, Loop Testing, Data Flow Testing, Mutation Testing

Unit VI	Test Automation	07 Hours

Automation and Testing Tools: Need for Automation, Categorization of Testing Tools, Selection of Testing Tools, Selection of Testing Tools, Costs Incurred in Testing Tools, Guidelines for Automated Testing, Overview of Some Commercial Testing Tools.

Testing Web-based Systems: Web-based System, Web Technology Evolution, Traditional Software and Web-based Software, Challenges in Testing for Web-based Software, Quality Aspects, Web Engineering, Testing of Web-based Systems

Text Books:

1. SOFTWARE TESTING Principles and Practices- Naresh Chauhan OXFORD UNIVERSITY
2. Software testing: Yogesh Singh, Cambridge University Press, First Edition

References Books:

1. Foundations of Software testing: Aditya P. Mathur, Pearson, Second Edition
2. Software Testing: Ron Patton, Pearson (SAMS), Second Edition
3. Software Quality , Mordechai Ben Menachem, Garry S. Marliss ,BS Publications

DKTES Textile and Engineering Institute, Ichalkaranji.
Final Year B. Tech. (Semester – VIII)
CSL458: Agile Methodology

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week Tutorials: 00Hrs/Week Practicals: 00 Hrs/Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Software Engineering

Course Outcomes :
On completion of the course, students will be able to

1. Describe basic concepts of agile methodology
2. Identify the roles and responsibilities in agile projects and their difference from projects following traditional methodologies.
3. Analyze the core practices behind several specific agile methodologies.

Course Contents

Unit I	Introduction	07 Hours

Need of Agile software development, agile context– Manifesto, Principles, Methods, Values, Roles, Artifacts, Stakeholders, and challenges. Business benefits of software agility.

Unit II	Project Planning	06 Hours

Recognizing the structure of an agile team– Programmers, Managers, Customers. User stories– Definition, Characteristics and content. Estimation– Planning poker, Prioritizing, and selecting user stories with the customer, projecting team velocity for releases and iterations.

Unit III	Project Design	06 Hours

Fundamentals, Design principles–Single responsibility, Open-closed, Liskov substitution, Dependency-inversion, Interface-segregation.

Unit IV	Design Methodologies	05 Hours

Need of scrum, Scrum practices –Working of scrum, Project velocity, Burn down chart, Sprint backlog, Sprint planning and retrospective, Daily scrum, Scrum roles– Product Owner, Scrum Master, Scrum Team.

Unit V	Extreme Programming	04 Hours

Core principles, values and practices. Kanban, Feature-driven development, Lean software development.

Unit VI	Testing	08 Hours

The Agile lifecycle and its impact on testing, Test driven development– Acceptance tests and verifying stories, writing a user acceptance test, Developing effective test suites, Continuous integration,

Code refactoring. Risk based testing, Regression tests, Test automation.

Text Books:

1. Ken Schawber, Mike Beedle, “Agile Software Development with Scrum”, International Edition, Pearson.
2. Robert C. Martin, “Agile Software Development, Principles, Patterns and Practices”, First International Edition, Prentice Hall.
3. Pedro M. Santos, Marco Consolaro, and Alessandro Di Gioia, “Agile Technical Practices Distilled: A learning journey in technical practices and principles of software design”, First edition, Packt Publisher.

References Books:

1. Lisa Crispin, Janet Gregory, “Agile Testing: A Practical Guide for Testers and Agile Teams”, International edition, Addison Wesley.
2. Alistair Cockburn, “Agile Software Development: The Cooperative Game”, 2nd Edition, Addison-Wesley

1. “Agile Software Development”, https://www.edx.org/course/agile-software-development
2. “Agile Software Development”, https://www.coursera.org/learn/agile-software-development Accessed on August 27, 2021.

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VIII)
CSL459: Human Computer Interaction

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week Tutorials: 00 Hrs/Week Practicals: 00 Hrs/Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: --

Course Outcomes:
On completion of the course, student will be able to–

1. Describe the basics of human-computer interaction.
2. Explain the principles to interaction design and different HCI models.
3. Illustrate various interactive application areas and their challenges.

Course Contents

Unit I	Introduction to HCI	06 Hours
Human: I/O channels – Memory – Reasoning and problem-solving; **The computer:** Devices – Memory – processing and networks; **Interaction:** Models – frameworks – Ergonomics – styles – elements – interactivity- Paradigms.		
Unit II	**Interaction Design**	**8 Hours**
Interaction Design basics – process – scenarios – navigation – screen design – Iteration and prototyping. HCI in software process – software life cycle – usability engineering – Prototyping in practice – design rationale. Design rules – principles, standards, guidelines, rules.		
Unit III	**Universal Design and Design Process Evaluation**	**07 Hours**
Universal Design: Universal Design Principles, Multi-modal interaction, Designing for diversity. **Design Process Evaluation:** What is evaluation?, Goals, evaluation through expert analysis, evaluation through user participation, choosing an evaluation method.		
Unit IV	**HCI Models and Theories**	**07 Hours**
Cognitive models: Goal and Task hierarchies**,** Linguistic models**,** The challenge of display-based systems, Physical and device models, cognitive architectures. **Socio-Organizational issues and stake holder requirements-** Organizational issues, capturing requirements.		
Unit V	**Models of the System**	**05 Hours**
Models of the System: standard formalism, interaction models, continuous behavior. **Modeling rich interaction:** Status-event analysis, Rich contexts, Low intension and sensor-based interaction.		
Unit VI	**Current and Future trends of HCI**	**06 Hours**
Groupware, Ubiquitous computing and augmented realities, Hypertext, Multimedia and WWW.		

Text Books:

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, “Human Computer Interaction”, 3rd Edition, Pearson Education.

References Books:

1. Interaction Design: Beyond Human-Computer Interaction 5th Edition by Helen Sharp, Jennifer Preece, Yvonne Rogers (Wiley).
2. New Directions in Third Wave Human-Computer Interaction: Volume 1 - Technologies Human-Computer Interaction Series) 1st ed. 2018 Edition by Michael Filimowicz, Veronika Tzankova (Springer).
3. Designing the User Interface: Strategies for Effective Human-Computer Interaction 6th Edition by Ben Shneiderman, Catherine Plaisant, Maxine Cohen, Steven Jacobs, Niklas Elmqvist, Nicholas Diakopoulos (Pearson).

Useful Links: --

1. https://www.hcibook.com/e3/

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VIII)
CSL460: Software Development Framework

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week Tutorials: 00Hrs/Week Practicals: 00 Hrs/Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Course Outcomes:
On completion of the course, student should be able to–

1. To describe fundamentals of application development framework such as EJB, struts, Spring and hibernate
2. To distinguish features of application development framework such as EJB, struts, Spring and hibernate.
3. To design application using application development framework such as EJB, struts, Spring and hibernate

Course Contents

Unit I	Java Beans and EJB	06 Hours

Java Bean- Concepts, Writing process, Applications

EJB Architecture: Overview of EJB software architecture, View of EJB Conversation, Building and Deploying EJBs, Roles in EJB

Unit II	EJB Applications	08 Hours

EJB Session Beans, EJB entity beans, Message Driven Bean, Lifecycle of Beans, EJB clients, Steps in developing an application with EJB.

Unit III	Struts 2: web application framework	07 Hours

what's a framework? Why use a framework? A brief history, Struts 2: the MVC pattern, How Struts 2 works, saying hello to Struts 2

Core concepts: Working with Struts 2 actions, Adding workflow with interceptors, Data transfer: OGNL and type conversion.

Unit IV	Building the view: tags and results	06 Hours

Building a view, UI component tags, Results in detail, exploring the validation framework, Extending Struts 2 with plug-ins: Plug-in overview, Common plug-ins, SiteMesh, Tiles, JFreeChart

Unit V	SPRING framework	07 Hours

Introduction to Spring, Spring Basic Concept, Spring Architecture, Features, Spring Advanced Concept, Spring MVC, Spring ORM

Unit VI	HIBERNATE framework	07 Hours

Introduction, Writing the application, application development approach, creating database and tables in MySQL, creating a web application, Adding the required library files, creating a java bean class, creating

hibernate configuration and mapping file, adding a mapping resource, creating JSPs.
Text Books:
1. Mastering Enterprise Java Beans-Ed Roman, John Wiley & Sons Inc. 2. Struts 2 in Action by Donald Brown, Chad Michael Davis, Scott Stanlick 3. Spring in Action 3rd edition, Craig walls, Manning Publication 4. Hibernate 2nd edition, Jeff Linwood and Dave Minter, Beginning Après publication
References Books:
1. Java Server Programming – Java EE7 (J2EE 1.7) Black Book. Kogent Solutions 2. Struts 2: The Complete Reference by James Holmes 3. Struts 2 Design and Programming: A Tutorial by Budi Kurniawan 4. Spring and Hibernate by Santosh Kumar K 5. Valesky – Enterprise Java Beans – Addison Wesley 6. Java Server Programming – Java EE7 (J2EE 1.7) Black Book. Kogent Solutions 7. Java EE 6 for Beginners, Sharanam Shah, Vaishali Shah, SPD

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. CSE (Semester – VIII)
CSL461: DevOps

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week Tutorials: 00Hrs/Week Practicals: 00 Hrs/Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Cloud Computing, Networking

Course Outcomes:
On completion of the course, student should be able to–

1. Describe basic concepts of devops and containerization
2. Apply devops tools for real time application development
3. Analyze devops applications

Course Contents

Unit I	Introduction	06 Hours

Introduction: - Understanding the principles and goals of DevOps, basics of Continuous Integration and Continuous Delivery, Benefits of DevOps, Overview of DevOps tools and technologies.

Unit II	Continuous Development	06 Hours

Introduction, What is Version Control system, Versioning, staging & un-staging, Branching, Merging, and rebase, Rollback, reset, Git ssh login, Installing Git on Windows,

Unit III	Introduction to Container	06 Hours

What are containers? What is Docker? Microservices, Docker installation and commands setup, Docker logs and Docker volumes, Building images, Docker Compose, Multistage Dockerfile

Unit IV	Continuous Integration using Jenkins	06 Hours

Introduction to Jenkins, How to change Home Directory, How to use Jenkins from the command line, How to create Users + Manage + Assign Roles, Jenkins authentication and authorization

Unit V	Continuous Delivery and Deployment	06 Hours

Implementing continuous delivery pipelines, Deploying applications to various environments (development, staging, production), Managing configuration and environment-specific variables

Unit VI	Cloud Computing and Infrastructure	06 Hours

Introduction to cloud computing platforms (e.g., AWS, Azure, Google Cloud), Provisioning and managing infrastructure in the cloud, Implementing auto-scaling and high availability

Text Books:

1. "DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology

Organizations" by Gene Kim, Jez Humble, Patrick Debois, and John Willis
References Books:
1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" by Gene Kim, Kevin Behr, and George Spafford
2. "Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation" by Jez Humble and David Farley

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. CSE (Semester – VIII)
CSL462 : Quantum Computing

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 03 Hrs /Week Tutorials: 00Hrs/Week Practicals: 00 Hrs/Week	03	MSE: 30 Marks ISE: 20 Marks SEE: 50 Marks

Pre-requisite: Parallel Computing Architecture

Course Outcomes:
On completion of the course, student should be able to–

1. Explain the implications of quantum computing on cryptography and security
2. State the four postulates of quantum mechanics and their application to computation
3. Analyze fundamental quantum algorithms
4. Describe the principles of quantum information and quantum communication

Course Contents

Unit I	Introduction to Quantum Computing	06 Hours

Introduction to Introduction. Information and Computations. Characteristics of Computational Systems. Computability and Algorithms. Computational Complexity. Quantum Computing. The Multiverse Interpretation of Quantum Mechanics.

Unit II	Mathematical Model of Quantum Computing.	06 Hours

Qubit. Qubit Measurement. Systems with Multiple Qubits. Measuring the Multiple Qubits Systems. Quantum System Evolution. Computations.

Unit III	Quantum Computer and Quantum Algorithms.	06 Hours

Deutsch's Problem. Quantum Computer Prototype, DIY. Quantum Computer Prototype. Solving the Deutsch's Problem. More Algorithms.

Unit IV	Shor's Algorithm.	06 Hours

Introduction, Factoring and the RSA. Factoring and Period Finding. Quantum Fourier Transform. Shor's Algorithm.

Unit V	Grover's Algorithm. A Quantum Computer Application Boundaries.	05 Hours

Intro. Grover's Algorithm. Grover's Algorithm. A Closer Look. Can we do Better than Grover? Are Quantum Computers Always Better?

Text Books:

1. P. Kaye, R. Laflamme, and M. Mosca. *An Introduction to Quantum Computing*. Oxford, 2007.
2. S. A. Fenner. *Course Notes* (https://cse.sc.edu/~fenner/csce790/notes/index.html). 2007, revised 2021 during the course.
3. M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information* (the 10th Anniversary Edition is optional but recommended). Cambridge University Press, 2000 (10th Anniversary Edition: 2010).

4. Parag K Lala, Mc Graw Hill Education, Quantum Computing, A Beginners Introduction, First edition (1 November 2020).
5. Michael A. Nielsen, Issac L. Chuang, Quantum Computation and Quantum Information, Tenth Edition, Cambridge University Press, 2010.
6. Chris Bernhardt, The MIT Press; Reprint edition (8 September 2020), Quantum Computing for Everyone.

References Books:

1. Michael A. Nielsen and Isaac L. Chuang. Quantum Computation and Quantum Information. Cambridge, UK: Cambridge University Press, September 2000. ISBN: 9780521635035.
2. Peres, Asher. Quantum Theory: Concepts and Methods. New York, NY: Springer, 1993. ISBN: 9780792325499.
3. Preskill, J. Notes on Quantum Computation.
4. V. Sahni, “Quantum Computing”, Tata McGraw-Hill Publishing Company, 2007.

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. CSE (Semester – VIII)
CSL463 : Full Stack Web Development Lab

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 02 Hrs./Week Tutorials: 00Hrs./Week Practicals: 02 Hrs./Week	03	CIE: 50 Marks SEE: 50 Marks

Pre-requisite: Computer Network, Web Technologies

Course Outcomes:

On completion of the course, student should be able to–

1. Explain the concepts of full stack web development.
2. Devise an architecture for solving the given problem.
3. Develop a web application using Angular and other back end technologies.

Course Contents

Unit I	Introduction to Typescript	03 Hours

Overview, Environment Setup, Basic syntax, Type, Variables, Operators, Decision Making, Function, Loops, Function, Numbers, String, Array, Tuples, Union, Interfaces, Classes, Objects, Namespaces, Modules, Ambients.

Unit II	Fundamentals of Angular	03 Hours

App, Routing, Managing Data, Form, Deployment, Environment Setup, Architectural Overview, Introduction to modules, Introduction to components, Introduction to services and DI

Unit III	Working with Angular- Part-I	06 Hours

Displaying data, Template syntax, User Inputs, Lifecycle hooks, Component interaction, component style, Angular element, Dynamic components, Attribute Directives and structural directives, Pipes.

Unit IV	Working with Angular- Part-II	06 Hours

Introduction, Reactive forms, Template Driven Forms, Form Validation, Dynamic Forms, Bootstraping, NgModules introduction, JSModules vs NgModules, Frequently used NgModules, Types of feature modules, Entry component, Feature Modules, Providers, Singaltone Services, Sharing NgModule, NgModule API, HTTP Client, Routing and navigation.

Unit V	NodeJS and ExpressJs	04 Hours

NodeJs architecture ,Modules: Built-in and custom, Event loop, Asynchronous application , Testing node application, Introduction to ExpressJs, Routing, Template engines, Middleware, Web Application components, Error handling, Testing application Express application

Unit VI	Mongodb	02 Hours

Relational vs NoSQL DB, MongoDB fundamentals, Data modeling , Aggregation pipeline, Grid FS, Performance optimization

Text Books:
1. MEAN Web Development by Amos Haviv, Packt publishing 2. Full Stack JavaScript Development with MEAN By Colin J. Ihrig, Sitepoint publishing 3. Getting MEAN with Mongo, Express, Angular, and Node by Simon Holmes, Manning Publishing
Web Links
1. https://www.tutorialspoint.com/typescript/ 2. https://www.w3schools.com/angular/default.asp 3. https://angular.io/ 4. https://www.w3schools.com/nodejs/default.asp 5. https://expressjs.com/ 6. https://www.mongodb.com/

DKTE Society's Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. CSE (Semester – VIII)
CSP464: Mobile Application Development Lab

Teaching Scheme:	Credits	Evaluation Scheme:
Lectures: 01 Hrs /Week Tutorials: 00 Hrs/Week Practicals: 02 Hrs/Week	02	CIE: - 50 Marks SEE: 50 Marks

Pre-requisite: Java Programming

Course Outcomes:
On completion of the course, student should be able to–

1. Explain fundamental concepts of Mobile application Development.
2. Design user interface for Mobile application.
3. Develop Mobile application using appropriate platform.

Course Contents

Unit I	Android Overview	04 Hours

Overview of Android, History, Android Versions, Android OS stack, Introduction to Android SDK, Android Emulator, Creating a Project, Project Directory Structure.

Unit II	Intents and Layouts	04 Hours

XML, Android View Hierarchies, Linear Layouts, Relative Layout, Table Layout, Frame Layout, Using Padding and Margins with Layouts. What Is Intent? Android Intent Messaging via Intent Objects, Types of Intents, Using Intents with Activities, Sending Intents (Telephony, SMS), Broadcast Receivers

Unit III	Input Controls, Input Events, Dialogs, Menus, Android Database	04 Hours

Buttons, Text Fields, Checkboxes, Radio Buttons, Toggle Buttons, Spinners, Event Listeners, Event Handlers, Dialogs: Alerts, Popups, Toasts, Menus, Options menu, Context menu, Popup menu, Handling menu click events, creating a Notification, Installing SQLite plugin, DbHelper, The Database Schema and Its Creation, Four Major Operations, Cursors, Example

Unit IV	iOS Basics	04 Hours

Introduction, XCode, iOS Simulator, Interface Builder, Understanding XCode, Accessing UI, Buttons, Views, Image Views, Switch and Text Field, Introduction, Print and comments, Variables and Primitives, Operators, Break, Control Flow, Collections, Dice Generator

Unit V	Intermediate UI	04 Hours

Introduction, Outlets and Constraints, Stack Views, Hard Coding Outlets, Segmented Controls, Break, View controller, Date Picker View, Picker View, Slides.

Unit VI	Login UI	04 Hours

Introduction, starting New App, Three Main components, Segmented Control, Text fields, Outlets in Code, Break, Login or Register Function, Revealing the Password, Constraints, User Notification
Text Books:
1. Beginning Android application development by Wei-Mag Lee 2. Learning Android by Marko Gargenta Publisher: O'Reilly Media 3. Swift: A Step-by-Step Guide for Absolute Beginners by Daniel Bell
References Books:
1. Application Development by Reto Meier Publisher: Wiley India
2. Android in Action Third Edition by W.Frank Ableson, Robi Sen, Chris King, C. Enrique Ortiz
3. Swift for Beginners (Develop and Design) by Boisy G. Pitre

List of Experiments **(It should consist of 10-12 experiments based on the following topics.)**	
1	Installation of Android SDK, emulator, Creating simple project and study of android project structure
2	Write a program to use of different layouts.
3	Write a program to study Intents for switching between activities.
4	Program to demonstrate Buttons, Text Fields, Checkboxes, Radio Buttons, and Toggle Buttons with their events handler.
5	Program to demonstrate Touch Mode, Menus with their events handler.
6	Write a program to study and use of SQLite database.
7	Introduction to XCode, iOS Simulator and Interface Builder
8	Program for Accessing UI Outlets, Buttons, Views and Image Views
9	Write a Program Intermediate UI using Swift-Outlets, Constraints, View controller and Data
10	Program to demonstrate Date Picker View, Picker View and Sliders
11	Create an iOS Application using Login Function, Register Function, Revealing password and User Notifications
12	Create an iOS Application using Scroll Views, Table views and Reusable Views

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VIII)
CSP465: Software Development Framework Lab

Lab Scheme:	Credits	Evaluation Scheme:
Practical: 02 Hrs. /Week	01	CIE: 50 Marks

Course Outcomes:

On completion of the course, student will be able to–

1. To describe fundamentals of application development framework such as EJB, struts, Spring and hibernate.
2. To design application using application development framework such as EJB, struts, Spring and hibernate.
3. To develop application using application development framework such as EJB, struts, Spring and hibernate.

List of Experiments
(It should consist of 10-12 experiments based on the following topics.)

1	Create a login Servlet application to authenticate the user based on username and password
2	Create a login JSP application to authenticate the user based on username and password
3	Study of Introduction and configuration of EJB.
4	Implementation of JavaBeans
5	Write a program to study use of Session Bean
6	Write a program to study use of Entity Bean
7	Write a program to study use of Message Driven Bean
8	Study of Introduction and configuration of struts 2
9	Write a MVC program for Login page with validation in struts 2
10	Introduction to hibernate framework and hello world example
11	Write a program to develop a web application supporting hibernate framework to perform database operation
12	Introduction to Spring framework and develop simple spring based application

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. CSE (Semester – VIII)
CSP466: DevOps Lab

Lab Scheme:	Credits	Evaluation Scheme:
Practical: 02 Hrs. /Week	01	CIE: 50 Marks

Course Outcomes:

On completion of the course, student will be able to–

1. Explain different DevOps Tools.
2. Build CI/CD pipeline for a web application.
3. Develop an application and deploy on Docker hub.
4. Build a web application and integrating DevOps tools.

List of Experiments

(It should consist of 10-12 experiments based on the following topics.)

1	Based on what you have learnt in the class, do the following steps: 1. Based on what you have learnt in the class, do the following steps: 2. Put the following files in the folder ● Code.txt ● Log.txt ● Output.txt 3. Stage the Code.txt and Output.txt files 4. Commit them 5. And finally push them to GitHub
2	Consider the Scenario ● Feature1 branch to be a public branch ● Feature2 branch to be a private branch. The company relies on a monolithic architecture and for now all the code resides in one file. The respective features have been added in the feature branches for file. Meanwhile, a security patch was made to the master branch, and now feature1 and feature2 branches are behind from master by 1 commit. **Tasks To Be Performed:** 1. Update Feature1 and Feature2 branch with the Security Patch. 2. Apply changes of Feature1 and Feature2 branches on master. 3. Finally push all the branches to GitHub. For solving this, please fork the repository to your GitHub account and then work. As a solution, please submit your GitHub's repository link.
3	Create the following scenario. Working Ubuntu Linux account with Git, JDK 11, Jenkins and maven installed. Working user on GitHub. Jenkins service to be up and running. Clone from the GitHub account. Push the Project in GitHub Account. Create the Jenkins Freestyle Job. Update the Git Repo Link. Build the Job & Run.
4	Installation of Docker & implementation of Docker commands. Create a Docker image of your Application and Deploy into Docker Hub.
5	Perform Containerization of using Docker and study of Docker Orchestration tools- Kubernetes, Docker Swarm etc.
6	Implement the following Task for implementation of Kubernetes. 1. Deploy a Kubernetes cluster for 3 nodes 2. Create a NGINX deployment of 3 replicas 3. Use the previous deployment. 4. Create a service of type Node Port for NGINX deployment. 5. Check the NodePort service on a browser to verify.

7	Implement the following Task for implementation of Kubernetes. 1. Use the previous deployment 2. Change the replicas to 5 for the deployment 3. Change the service type to Cluster IP 4. Deploy an NGINX deployment of 3 replicas 5. Create a NGINX service of type Cluster IP
8	Create a Web based Application as a Calculator with Kubernetes and deployment.
9	Install Terraform & Perform basic Terraform Operations
10	Deploy End-to –End Architecture of Terraform on AWS
11	Study & Implementation of AWS- DevOps tools- AWS Code Pipeline, AWS Code Build, AWS Code Deploy, AWS Code Star and Aws Lambda.
12	Capstone Course Project with AWS & DevOps.

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. AI (Semester – VIII)
CSP467 : Quantum Computing Lab

Teaching Scheme:	Credits	Evaluation Scheme:
Practicals: 02 Hrs/Week	01	CIE: 50 Marks

Course Outcomes:
On completion of the course, student should be able to:

1. Analyze the computation models.
2. Model the circuits using quantum computation, environments and frameworks.
3. Evaluate the quantum operations such as noise and error–correction.

List of Experiments

1. Single qubit gate simulation - Quantum Composer
2. Multiple qubit gate simulation - Quantum Composer
3. Composing simple quantum circuits with q-gates and measuring the output into classical bits.
4. IBM Qiskit Platform Introduction
5. Implementation of Shor's Algorithms
6. Implementation of Grover's Algorithm
7. Implementation of Deutsch's Algorithm
8. Implementation of Deutsch-Jozsa's Algorithm
9. Integer factorization using Shor's Algorithm
10. QKD Simulation
11. Mini Project such as implementing an API for efficient search using Grover's Algorithms

DKTES Textile and Engineering Institute, Ichalkaranji
Final Year B. Tech. (Semester – VIII)
CSD468: Project-II

Lab Scheme:	Credits	Evaluation Scheme:
Practical: 08 Hrs. /Week	08	CIE: 50 Marks SEE: 50 Marks

Pre-requisite: Software Engineering

Course Outcomes:

On completion of the course, a team of student will be able to–

1. Update the requirement specifications and design document of their project
2. Write code according to the structure specified in SRS.
3. Carry out unit as well as integrated testing
4. Write a report covering details of the project.
5. Give presentation on project

Project Guidelines

The group will continue to work on the project selected during the semester VII and submit the completed project work to the department at the end of semester VIII as mentioned below.

The workable project.

The project report in the bound journal complete in all respect with the following : -

i. Problem specifications

ii. System definition – requirement analysis.

iii. System design – dataflow diagrams, database design

iv. System implementation – algorithm, code documentation

v. Test results and test report.

vi. In case of object oriented approach – appropriate process be followed.

CIE will be jointly assessed by a panel of teachers appointed by head of the department. SEE examination will be conducted by internal and external examiners as appointed by the COE.

Note:

1. Project work should be continually evaluated based on the contributions of the group members, originality of the work, innovations brought in, research and developmental efforts, depth and applicability, etc.
2. Two mid-term evaluations should be done, which includes presentations and demos of the work done.
3. Care should be taken to avoid copying and outsourcing of the project work.

The Continuous internal evaluation (CIE) is to be done as follows.

1. Project Evaluation I – 15 marks.
2. Project Evaluation II – 15 marks.
3. Project Internal – 20 marks.

Semester End Exam :

The semester end exam will be conducted by examiners appointed by exam coordinator.